WEEK 1 REPORT

TEAM FORMATION & DATA HUNTING

MONEYBALL – IPL PLAYER VALUATION

DATA SCIENCE AND VISUALIZATION | COURSE CODE: 24ADI204

DATA-VIZ HACK-A-BIT (2026–2027)

TEAM NUMBER: 05

Team Member	Register Number
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Prepared from the Week 1 Google Colab notebook, the team details slide and the latest notebook screenshot supplied by the team.

1. Introduction

Moneyball – IPL Player Valuation is a data science project that explores Indian Premier League (IPL) player performance information alongside available auction-related information. The project aims to examine how player statistics relate to player valuation and to present insights through an interactive dashboard in later stages.

Week 1 establishes the project foundation. The team loads the prepared merged IPL master dataset into Google Colab, inspects its structure and reviews initial records and descriptive information. This report documents the activities evidenced in the supplied notebook.

2. Problem Statement

IPL player records contain multiple performance measures and descriptive or auction-related attributes. Before meaningful analysis can be performed, the dataset must be loaded and its structure understood. Week 1 addresses this initial requirement through basic dataset loading and inspection.

3. Objectives

• Load the prepared IPL master CSV into Google Colab.
• Inspect sample records, dimensions, column names and data types.
• Generate initial numerical summaries using Pandas.
• Demonstrate basic numerical operations using NumPy.
• Establish a baseline for data-quality checks, analysis and dashboard development in later weeks.

4. Team Formation

The team number and member details are taken from the team-members slide supplied for this project.

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5. Dataset Selection and Description

The executed notebook loads **IPL_Moneyball_Merged_2008_2026.csv**. The notebook output reports 3,311 rows and 27 columns. The fields cover season and player information, batting and bowling performance, team/player descriptors, acquisition details and price-related information.

Dataset property	Observed value
Rows / records	3,311
Columns / attributes	27
Distinct player values	794
Distinct seasons	19
Season range	2008–2026

5.1 Dataset fields

The 27 columns reported in the notebook are: season, player, runs, balls_faced, innings, fours, sixes, dismissals, balls_bowled, runs_conceded, wickets, matches_bowled, auction_name, team, role, nationality, status, acquisition_type, base_price_lakh, final_price_lakh, bat_style, bowl_style, player_full_name, strike_rate, batting_avg, economy and bowling_avg.

5.2 Dataset merge/source details

The notebook reads the already-merged master CSV. It does not show the original source CSV filenames, source URLs, merge code, join keys or the detailed merging procedure. These details are not inferred from the final merged file.

6. Tools and Libraries

Pandas: Used to read the CSV into a DataFrame and inspect tabular data using read_csv(), head(), tail(), sample(), shape, columns, info(), describe(), dtypes and nunique().

NumPy: Used to create a numerical array and calculate its sum, mean, maximum and minimum.

Matplotlib and Seaborn: Imported for visualization work. The supplied Week 1 notebook evidence does not show a graph generated using these libraries.

7. Methodology and Implementation

Step 1 – Import libraries

The notebook imports pandas as pd, numpy as np, matplotlib.pyplot as plt and seaborn as sns. The displayed output confirms that all libraries were imported successfully.

Step 2 – Load the master dataset

The dataset is loaded using pd.read_csv('IPL_Moneyball_Merged_2008_2026.csv'). The notebook prints “Dataset loaded successfully!”, confirming that the file was read in the Colab session.

Step 3 – Preview records

head() displays the first five records, tail() displays the last five records and sample(5) displays five randomly selected records. These previews provide an initial view of the data.

Step 4 – Inspect dimensions and columns

The shape output is (3311, 27), meaning 3,311 rows and 27 columns. The columns operation displays the field names listed in Section 5.1.

Step 5 – Inspect DataFrame information and data types

The info() output reports 3,311 entries with index values from 0 to 3310. It reports 11 integer columns, 6 floating-point columns and 10 object columns. The displayed memory usage is approximately 698.5 KB. The dtypes operation shows the data type of each field.

The info() output also shows incomplete fields. Examples visible in the notebook include role with 893 non-null values, nationality with 1,015, base_price_lakh with 202 and final_price_lakh with 1,831. These are initial data-quality observations; this Week 1 notebook does not perform missing-value treatment.

Step 6 – Generate descriptive statistics

describe() generates summary statistics for numeric columns. In the recorded output, runs has a mean of approximately 111.74, median of 34 and maximum of 973. final_price_lakh has 1,831 numeric observations, a mean of approximately 366.02 and a maximum of 2,520.

Step 7 – Demonstrate NumPy operations

The notebook creates the sample array [450, 520, 380, 600, 700]. The displayed results are sum = 2,650, mean = 530, maximum = 700 and minimum = 380. This is a separate numerical demonstration, not a calculation over the IPL dataset.

Step 8 – Count distinct players and seasons

The notebook uses df['player'].nunique() and df['season'].nunique(). The outputs are 794 distinct values in the player column and 19 distinct season values. The dataset shape is also printed as (3311, 27).

8. Results and Initial Observations

- The master dataset contains 3,311 records and 27 columns, exceeding the project's minimum requirement of 1,000 rows.
- The dataset covers 19 seasons, with the notebook output indicating 2008–2026.
- It includes batting and bowling measures such as runs, wickets, strike_rate and economy, along with team, role, acquisition_type and price-related fields.
- Some columns contain missing values and should be examined during later data-quality and cleaning activities.
- The NumPy calculation is a demonstration using a separate sample array.

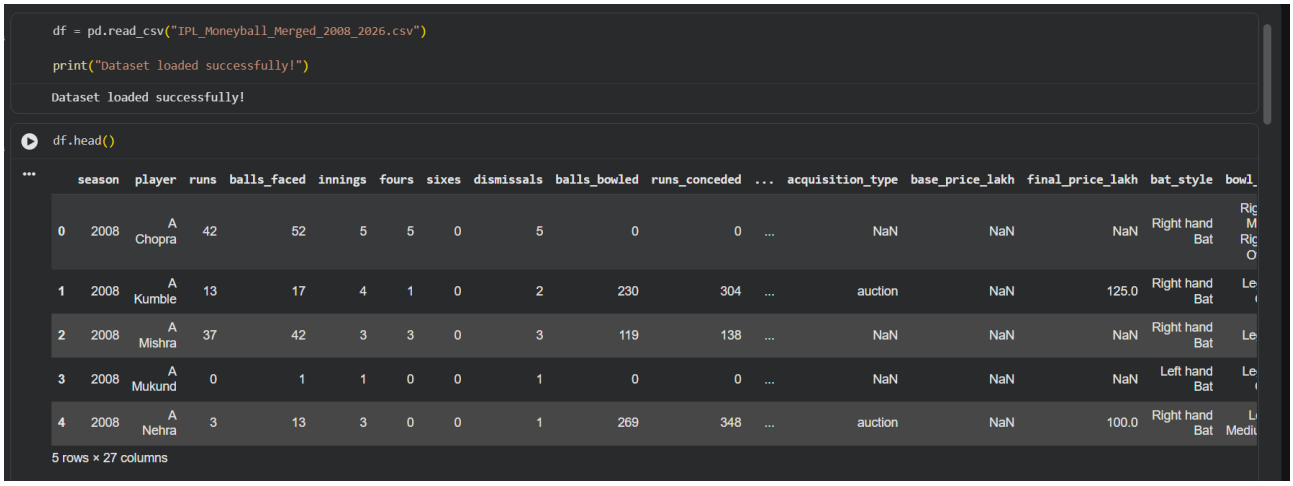
9. Before-and-After / Baseline Comparison

Week 1 is the initial dataset loading and inspection stage. The uploaded notebook does not show data cleaning, missing-value imputation, duplicate removal, outlier treatment, scaling, encoding, feature selection or dashboard transformation. Therefore, no processed after-state is claimed for this week.

Measure	Week 1 baseline
Dataset shape	3,311 rows × 27 columns
Distinct player values	794
Distinct seasons	19
Missing values	Present in multiple columns; not treated in Week 1
Transformations shown	No cleaning/transformation shown in supplied notebook

10. Google Colab Evidence

The screenshot below shows the executed dataset-loading cell, the successful loading message and the first five rows of the merged IPL dataset, with the output indicating 5 rows × 27 columns.



```
df = pd.read_csv("IPL_Moneyball_Merged_2008_2026.csv")

print("Dataset loaded successfully!")
```

```
Dataset loaded successfully!
```

```
df.head()
```

	season	player	runs	balls_faced	innings	fours	sixes	dismissals	balls_bowled	runs_conceded	...	acquisition_type	base_price_lakh	final_price_lakh	bat_style	bowl_
0	2008	A Chopra	42	52	5	5	0	5	0	0	...	NaN	NaN	NaN	Right hand Bat	Rig M Rig O
1	2008	A Kumble	13	17	4	1	0	2	230	304	...	auction	NaN	125.0	Right hand Bat	Le
2	2008	A Mishra	37	42	3	3	0	3	119	138	...	NaN	NaN	NaN	Right hand Bat	Le
3	2008	A Mukund	0	1	1	0	0	1	0	0	...	NaN	NaN	NaN	Left hand Bat	Le
4	2008	A Nehra	3	13	3	0	0	1	269	348	...	auction	NaN	100.0	Right hand Bat	L Mediu

5 rows × 27 columns

Figure 1. Latest Google Colab screenshot supplied for Week 1 dataset loading and preview.

11. Team Contributions

The supplied team-members slide confirms the five team members. Specific individual task allocations were not provided in the supplied material, so individual contributions are not assigned in this report.

12. Conclusion

Week 1 established the project team and the initial data workflow for Moneyball – IPL Player Valuation. The prepared master dataset was loaded successfully into Google Colab and inspected using Pandas. The notebook verified 3,311 rows, 27 columns, 794 distinct player values and 19 distinct seasons. NumPy operations were demonstrated using a sample array. The dataset baseline and initial missing-value observations provide a starting point for subsequent inspection, cleaning, analysis and dashboard development.

13. References and Project Records

1. Week 1 Google Colab notebook: Week01_Team_Formation_and_Data_Hunting.ipynb (team project repository).
2. Master dataset used in the notebook: IPL_Moneyball_Merged_2008_2026.csv (team-provided file).
3. Original dataset source URLs and original source dataset citations should be added from the team's dataset collection records.

14. Final Review Checklist

- Confirm the exact original dataset filenames, source URLs and merge keys from the original preparation process.
- Keep the same master dataset for subsequent project weeks, as required by the project plan.